

$$5.2) A = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}_{3 \times 3} \quad \begin{cases} d_1 = 2 \\ d_2 = 3 \end{cases} \quad \begin{aligned} \Lambda_{d_1} &= \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - y + z = 0 \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\} \\ \Lambda_{d_2} &= \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \end{aligned}$$

• podle nápovědy zvolíme bázi, nejprve bude volit bázi z vlastních vektorů

$$B = \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right)$$

$$\rightarrow \text{podle tvrzení 9.56: } [A]_B^B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\rightarrow \text{podle tvrzení 9.60: } [A]_K^K = [id]_K^B [A]_B^B [id]_B^K \quad \text{kde } [id]_K^B = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array} \right) \sim$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array} \right) \quad [id]_B^K = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$[A]_K^K = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$\text{Hledaná matice } A \text{ je tedy } A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$$
